

Magnetic Matter Movers Build Sheet 002 — The Smallest Beamed-Thermal Rocket

A water droplet, levitated on sound, propelled by light

Phase 0 of the launch-architecture ladder. Same physics as the flight vehicle with zero substitutions: dumb mass + energy at the point of use + steam thrust. Also the strangest catalog piece.

1. What This Is

A ~2 mm dyed-water droplet floats in an acoustic levitation trap (TinyLev, the open-source cousin of Bristol’s MATD). A laser fired **horizontally** at the droplet evaporates its lit side; the departing vapor’s momentum recoils the droplet **away from the beam** — vapor-recoil propulsion, the identical mechanism as the launch vehicle’s heat-exchanger stage. The trap doubles as the instrument: it’s a spring, so the droplet’s displacement from the trap node is a direct thrust readout. A microthrust balance on a desk.

Two facts that make it more than a toy: - At 500 mW absorbed, thrust exceeds droplet weight (**T/W = 1.2**) — the droplet could hover on its own steam - Demo flux is 30-160 kW/m² vs. the flight panel’s 48 MW/m² — same mechanism, 1,000× below flight conditions

2. Predicted Performance (validate against these)

2 mm droplet: 4.2 mg, weight 41 µN. Table is **absorbed** power; india-ink-dyed water absorbs ~50-80% of incident, so incident laser power runs ~1.5-2× these values.

Absorbed	Flux on droplet	Evap rate	Droplet life	Thrust	T/W	Trap displacement*
5 mW	1.6 kW/m²	2.2 µg/s	~30 min	0.5 µN	0.01	0.05-0.5 mm (camera only)
50 mW	16 kW/m²	22 µg/s	~3 min	5.0 µN	0.12	0.5-5 mm
100 mW	32 kW/m²	44 µg/s	~95 s	10 µN	0.24	1-10 mm (naked eye)
500 mW	159 kW/m²	0.22 mg/s	~19 s	50 µN	1.21	ejection likely

*TinyLev-class trap stiffness ≈ 1-10 µN/mm; calibrate yours in §7 step 2.

Predicted dynamics to look for: (a) steady offset along the beam axis at low power; (b) at higher power, a **limit-cycle flicker** — pushed off-beam → thrust dies → spring pulls it back into the beam → pushed again (beam-riding oscillation, a real flight-control phenomenon); (c) the endgame — as the droplet evaporates, T/W rises and it gets livelier before vanishing.

3. Bill of Materials (~\$150)

Item	Spec	Est.
TinyLev acoustic levitator	open-source design: 72× 16 mm 40 kHz transducers, Arduino Nano, L298N driver, 3D-printed frame — kit or self-build	\$70
Laser module — starter	≤5 mW, 650 nm red, Class 3R, dot focusable	\$10
Laser module — main	100-500 mW, 445-450 nm, adjustable focus, TTL/PWM power control	\$30
Safety goggles	OD 4+ at 445-455 nm, CE/ANSI rated — for every person in the room	\$30
Enclosure	cardboard/foamcore box, matte-black interior; black felt beam dump	\$10
Droplet handling	1 mL syringe + blunt 25 ga needle	\$5
Dye	india ink (best) or black food coloring	\$3
Camera	phone with macro/slow-mo on a small tripod, mm-grid card behind trap	on hand
Power meter (optional)	cheap laser power meter for absorbed-power calibration	\$25
Photodiode (optional)	any photodiode + resistor into a multimeter/scope, aimed at the droplet from the side — backscatter is a second data channel (evaporation events show as scatter flickers)	\$2

4. Safety — read before powering anything

Laser (the real hazard): - 100+ mW at 450 nm is **Class 3B: a stray reflection causes permanent eye damage faster than you can blink**. The droplet is a spherical mirror that sprays specular reflections in ALL directions. - Non-negotiables: rated goggles on before power on; beam path fully inside the enclosure; beam terminates in the dump; remove watches/rings; beam horizontal at desk height

never at face height; key/switch control so it can't power on by accident. - **Graduation rule: run the entire §7 protocol at ≤5 mW red first.** Camera-tracked displacement at 0.5 μN is a complete, publishable measurement. Move to Class 3B only after the low-power protocol is routine.

Ultrasound: TinyLev runs high-intensity 40 kHz — inaudible to you, unpleasant for pets (cats hear well past 40 kHz). Keep animals out of the room; keep fingers out of the trap while loading (tingling is normal, prolonged exposure isn't).

5. Build Sequence

1. Assemble TinyLev per the open-source guide; verify all transducers phase-matched (the guide's oscilloscope or buzz test). Confirm it traps a 1-2 mm foam bead rock-solid before ever trying liquid.
2. Mount trap inside the enclosure, mm-grid card behind the trap axis, camera port on one side, laser port on the adjacent side so the beam runs **horizontal**, through a trap node, into the felt dump.
3. Rigidly mount the starter (red, ≤5 mW) laser; align to the node using a trapped foam bead as the target. Lock it down.
4. Mix dye: ~1 drop india ink per mL water. Darker = more absorption = more thrust per incident mW.

6. Droplet Loading (the finicky part)

1. Trap a foam bead first to confirm the trap is live; remove it.
2. Draw dyed water; touch a small hanging drop from the needle tip *slowly* into a pressure node — the trap will pluck it off. Start small (~1 mm); oversized droplets flatten, wobble, and atomize.
3. If droplets keep bursting: smaller droplet, or reduce drive amplitude slightly. Stable droplets sit still, slightly oblate.

7. Measurement Protocol

1. **Baseline:** trapped droplet, laser off. Record 30 s. Note rest position on the grid.
2. **Calibrate trap stiffness:** tilt the whole enclosure by a measured small angle θ ; the droplet's weight component $mg \cdot \sin\theta$ is a known force; its displacement gives k (μN/mm). Two or three angles → a line.
3. **Thrust runs:** laser on at fixed power, record until steady offset (or oscillation), laser off, watch it snap back. Repeat at 3-5 power levels. Slow-mo the higher-power runs.
4. **Reduce:** displacement $\times k$ = thrust. Plot thrust vs. incident power — prediction is a straight line whose slope gives your absorbed fraction. Photograph the droplet diameter each run (shrinkage → evaporation rate → cross-check via the table).
5. **The money shots:** (a) horizontal displacement with beam horizontal — proof it's recoil, not convection (convection pushes up, recoil pushes along the beam); (b) the snap-back on laser-off; (c) the limit-cycle flicker if you find its power threshold.
6. **Beam-riding offset map (imported from the 28 GHz gyrotron program's misalignment studies):** at fixed power, park the beam deliberately off the droplet's rest position in ~0.5 mm steps (laser on a micrometer stage or shim the mount). Record steady-state displacement at each offset → force-vs-offset curve. This is the bench-scale version of published beam-riding stability data: it tells you whether an off-axis vehicle gets pushed back toward the beam (self-centering) or shoved further out (divergent) — the core control question for any beamed-flight system. If the photodiode is fitted, log its channel too; scatter amplitude vs. offset is your transmitter-as-sensor demonstration.

8. Failure Modes

Symptom	Cause	Fix
Nothing happens	Undyed/underdyed water — beam passes through	More ink
Droplet atomizes on laser-on	Boiling onset / trap drive too hot	Lower power; smaller droplet; reduce drive
Droplet drifts without laser	Room air currents	Enclosure sealed; let it settle 60 s
Displacement vertical not horizontal	You're seeing convection or beam misaligned above/below node	Re-align beam through node center
Droplet ejects at high power	T/W approaching 1 — working as designed	Celebrate, then lower power for data runs

9. Where This Goes

- **Lab note 001:** thrust-vs-power curve + the horizontal-displacement proof = the first Magnetic Matter Movers publication, and the cheapest credible evidence of the architecture's core mechanism
- **v2:** swap dyed water for a black-anodized aluminum micro-panel with a wet face — a heat-exchanger *shape* instead of a bare droplet; measure thrust vs. panel angle
- **Catalog piece:** enclosure with a window, one-button run, the two curves printed on the base — *"the smallest beamed-thermal rocket in existence"*

Companion documents: Build Sheet 001 (Inductrack, Phase 0); Space Transportation Architecture (§Demonstration Ladder, §Technology Risk Register — heat exchanger).